Unit 5 Exponential and Logarithmic Functions

Basic

1. Find the values of x and y which satisfy the equations

$$3^{x+y} = \sqrt[3]{27},$$
$$\frac{4^y}{2^x} = \left(\frac{1}{2}\right)^{-3}.$$

[4]

[N19/I/10(a)]

Intermediate

2. In 2010, there were an estimated 2400 birds of a particular species in a certain country. In 2013, the numbers were estimated to have fallen to 2050. Scientists believe that the number of these birds, N, can be modelled by the formula $N = 2400e^{-kt}$, where t is the time in years after 2010.

   (a) Estimate the number of birds in the year 2017. [4]

   (b) The species is labelled "under threat" when the number of birds falls below 1200.
   Estimate the year in which the species might be first labelled "under threat". [2]

[N22/II/1]

3. Solve the equation $2^x + 4^{x-1} = 15$. [5]

[N22/II/6(a)]

4. **(a)** Show that the solution of the equation $6^x = 5 \times 3^{x+1}$ is $x = \dfrac{\lg 15}{\lg 2}$. [4]

   (b) Express the equation $\log_3 x + \log_9 (x + 2) = 2$ as a cubic equation in x. [4]

[N21/I/12]

5. Show that the equation $3e^x + 5 = 2e^{-x}$ has only one solution and find its value correct to 2 significant figures. [5]

[N21/II/1]

6. **(a)** Given that $\left[\left(\dfrac{50}{3}\right)^{-2} \times \sqrt{3^3}\right] \div \dfrac{5}{2} = 2^a\, 3^b\, 5^c$, find the value of each of a, b and c. [3]

   (b) The value of a painting increases by 7% each year. Given that its value at the beginning of 2014 was \$1 000 000, find its value, to 2 significant figures, at the beginning of 2020. [2]

[N20/I/2]

7. **(a)** Solve the equation $e^x(1 + e^x) = \dfrac{3}{4}$. [3]

   (b) Solve the equation $1 + \log_2 y + \dfrac{1}{\log_8 2} = \log_2 (y + 3)$. [4]

   (c) In order to obtain a graphical solution of the equation $x = \ln\left\{\left(\dfrac{2x+7}{3}\right)^2\right\}$ a suitable straight line can be drawn on the same set of axes as the graph of $y = 3e^{\frac{x}{2}} + 4$. Make $e^{\frac{x}{2}}$ the subject of $x = \ln\left\{\left(\dfrac{2x+7}{3}\right)^2\right\}$ and hence find the equation of this line. [3]

[N20/II/8]

8. (i) A manufacturer produces a disinfectant that destroys 21% of all known germs within one minute of use. If N is the number of germs present when the disinfectant is first used, and assuming germs continue to be destroyed at the same rate, explain why the number of germs expected to be alive after n minutes is given by $(0.79)^n N$. [2]

(ii) The manufacturer decides to advertise by stating that the disinfectant destroys x% of all known germs within 20 minutes of use. Calculate, to 2 significant figures, the value of x. [2]

(iii) Given that the number of germs expected to be alive after n minutes can be expressed as Ne^{kn}, find the value of the constant k. [2]

[N19/I/5]

9. Given that $\sqrt{125^x} = \dfrac{5^{1-x}}{25}$, find the value of $\sqrt{125^x}$. [4]

[N18/I/1]

10. (i) Show that $\log_3 x + \log_9 x = \dfrac{3 \lg x}{2 \lg 3}$. [3]

(ii) Hence solve the equation $\log_3 x + \log_9 x = 4$. [2]

[N18/I/6]

11. (i) Solve the equation $\log_5 (x-1) - \log_5 (x+1) = 1 + \log_5 \dfrac{1}{7}$. [4]

(ii) Solve the equation $\log_y 100 = \lg y$, giving your answer to 2 significant figures. [5]

[N17/II/5]

12. (a) The percentage, P, of carbon-14 remaining in a piece of fossilised wood is given by $P = 100e^{-kt}$, where k is a constant and t is measured in years. It takes 5730 years for the carbon-14 to be reduced to half of the original amount. Calculate

(i) the value of k, [3]

(ii) the percentage of carbon-14 which would indicate a fossil age of 8000 years. [2]

(b) The size, S, and intensity, I, of a naturally occurring event are connected by the formula $S = \lg \dfrac{I}{c}$, where c is a constant. Calculate, to 1 decimal place, the size of the event which has intensity 50 times that of an event of size 2.4. [4]

[N17/II/7]

13. (i) Given that $u = 2^x$, express $2^{2x-1} = 2^{x+2} - 6$ as an equation in u. [3]

(ii) Hence find the values of x for which $2^{2x-1} = 2^{x+2} - 6$, giving your answer, where appropriate, to 1 decimal place. [4]

(iii) Explain why the equation $2^{2x-1} = 2^{x+2} - k$ has no solution if $k > 8$. [3]

[N16/II/7]

14. The graph of $y = \log_a x$ passes through the points with coordinates $(8, 3)$, $(1, b)$ and $(c, -2)$.

(i) Determine the value of each of the constants a, b and c. [3]

(ii) Sketch the graph of $y = \log_a x$. [2]

[N15/I/2]

15. The number of bacteria in a culture doubles every 3 hours. It is given that N_0 is the number of bacteria present at a particular time and that N is the number of bacteria present t hours later. Calculate the value of the constant k in the relationship $N = N_0e^{kt}$. [4]

[N15/I/3]

Advanced

16. (a) Solve the equation $\log_2 x + \log_{16} x = -2.5$. [3]

(b) It is given that $\lg z - \lg y = \lg (z + y)$.

(i) Express z in terms of y. [3]

(ii) State the range of values of z and explain clearly why $0 < y < 1$. [2]

[N19/II/4]